FLUID AUTOMATION

Project reports
a.y. 2023/2024

GROUP 17

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POLITECNICO DI TORINO

FLUID AUTOMATION



POLITECNICO
DI TORINO

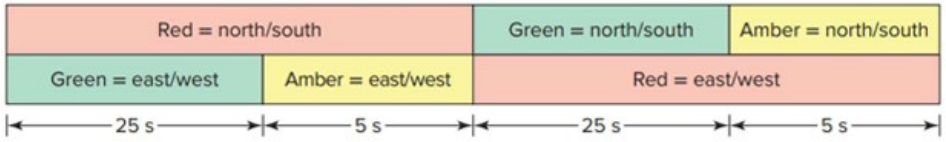
Task 1

CROSS INTERSECTION

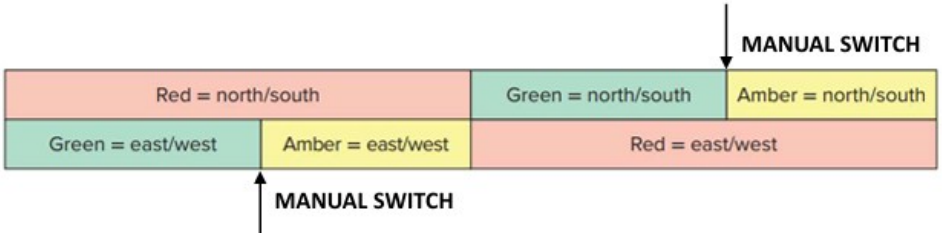
Objective, explanation

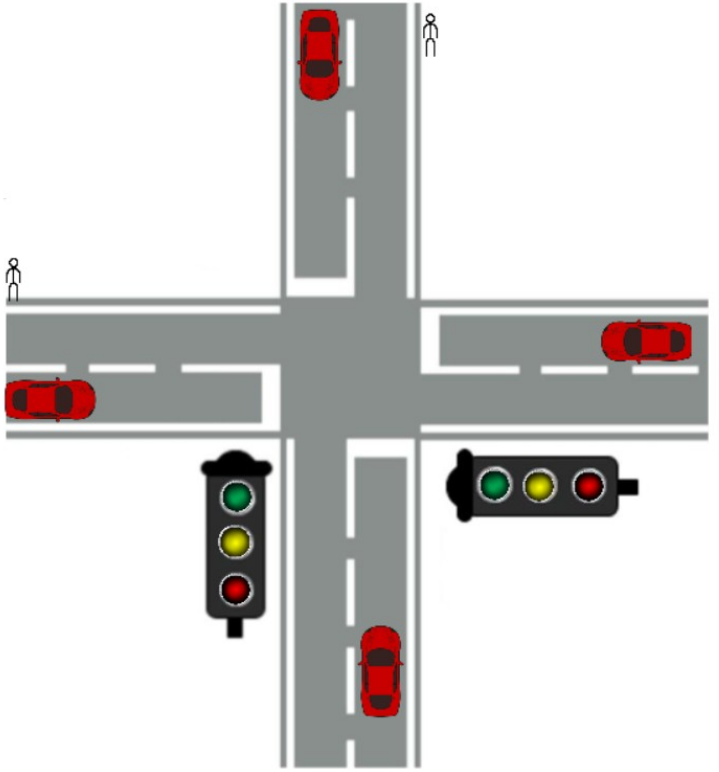
The aim of this project is to design a logic to manage a cross intersection using traffic lights. The system has three modes of operation:

1. **OFF:** all the traffic signals blink yellow with a specific pe
   When the intersection is in the "OFF" mode, the operato
   shall be able to reconfigure the lengths of the green anc
   red lights, by setting the desired times in the control par
2. **NORMAL:** the traffic signals follow the timing diagram
   below, in an automatic fashion



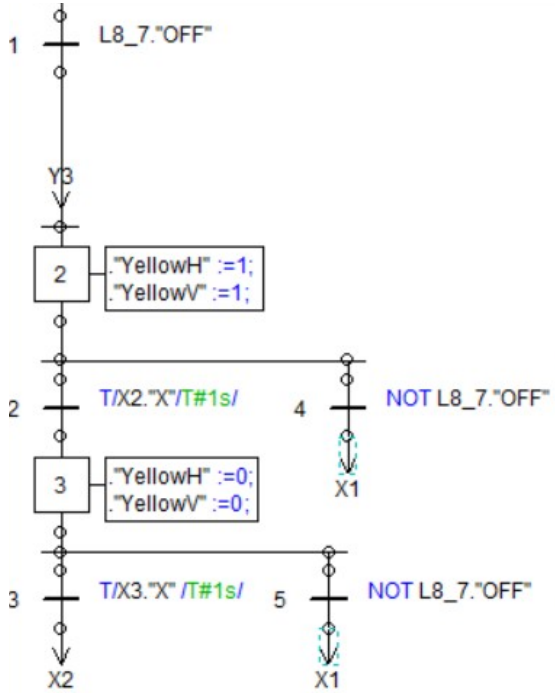
3. **MANUAL:** the operator shall be able to access the manu
   mode via a pushbutton in the control panel, and manuall
   switch the direction of traffic flow via a second pushbut

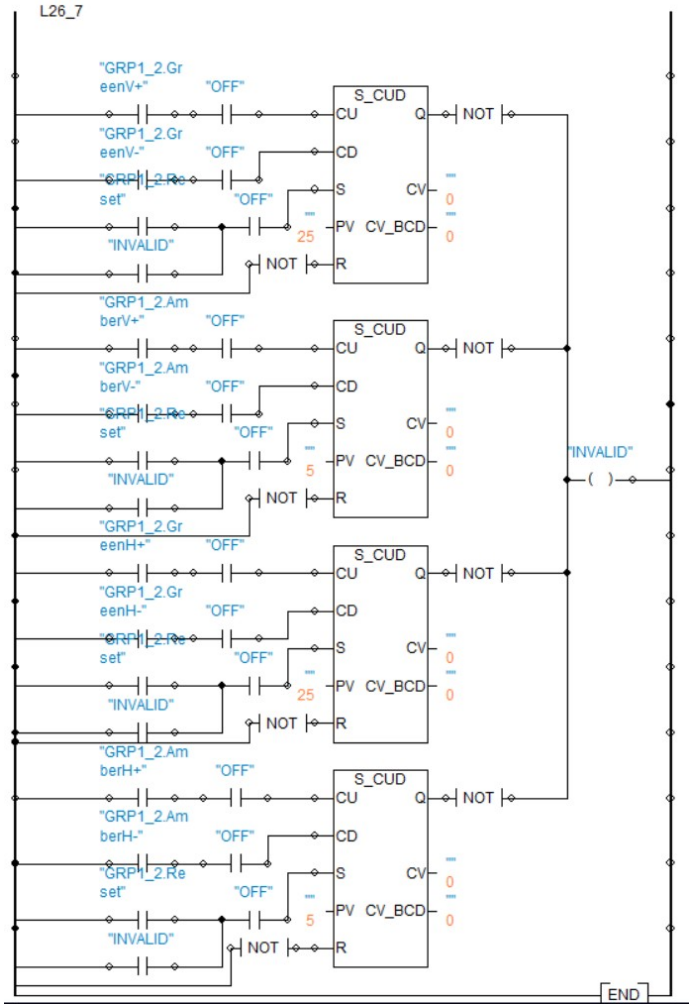




OFF MODE

In the **OFF** mode, all the traffic lights must blink yellow with a frequency equal to 0.5 Hz in our case as we can see in the Grafcet on the left.
In addition, the operator can change the length of the red and green lights which is guaranteed in the ladder code on the right

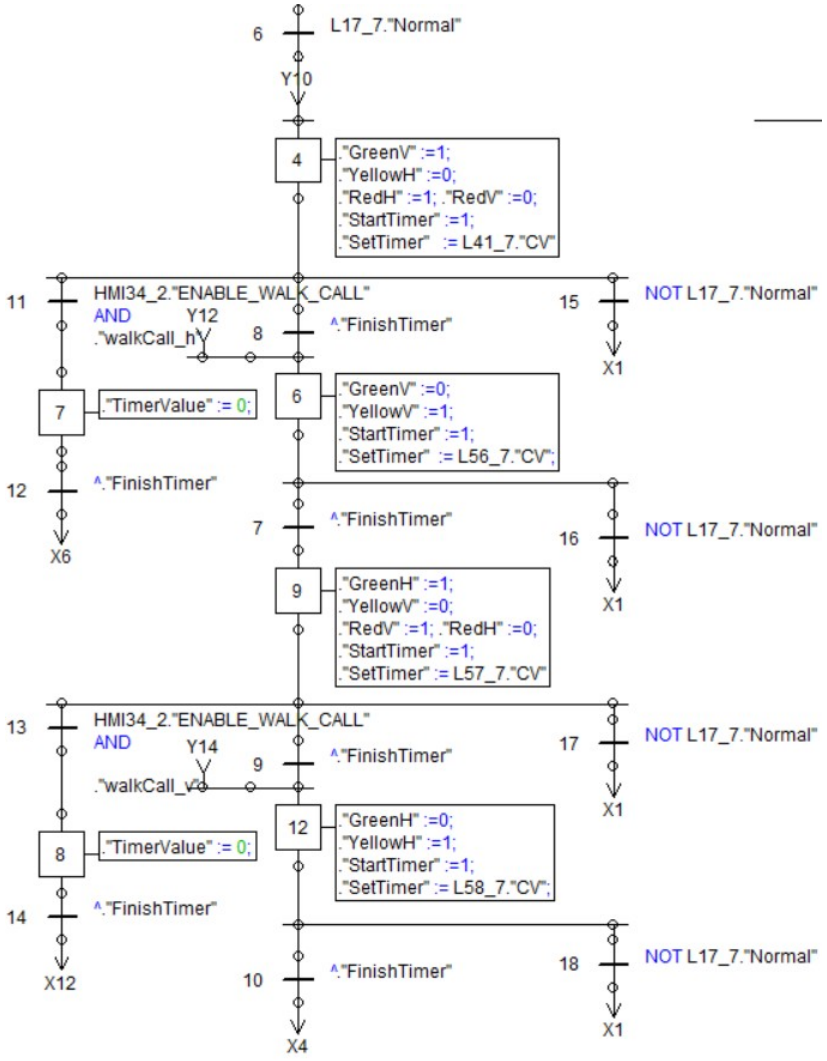




NORMAL MODE

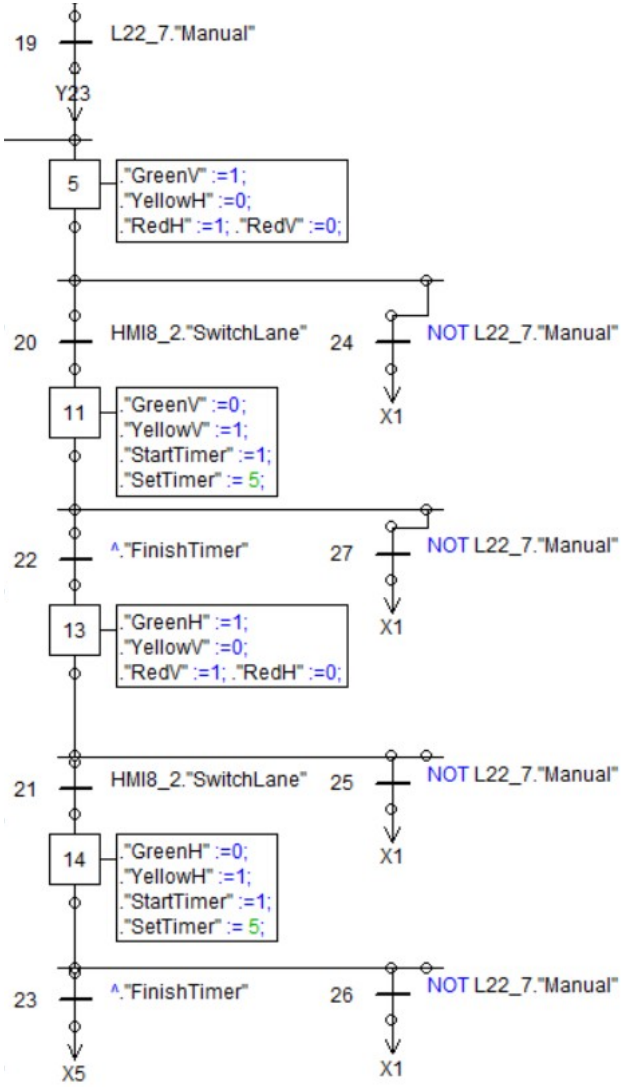
In the **NORMAL** mode, the cycle (green → yellow → red) should repeat continuously alternating between the vertical and horizontal lane, respecting the deadlines for the red and green lights set in the OFF mode.

In addition to the requirements, the green time can be skipped if a pedestrian of the opposite lane has arrived (virtually pressed the call button), with an option to enable or disable this functionality.



MANUAL MODE

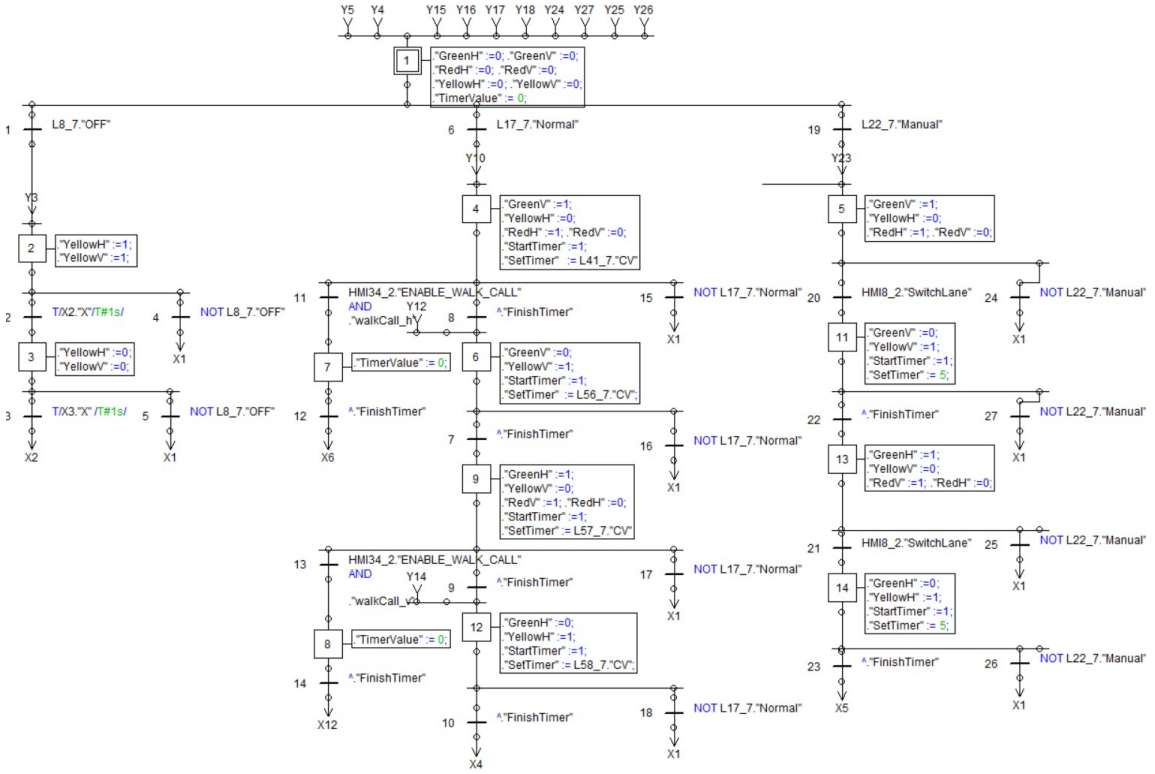
In the manual mode, the concept of switching lights is the same, except that every change (green → red) is triggered by a push button from the user.



MAIN GRAFCET

All the abovementioned modes are managed by a grafcet code which ensures the correct functioning of each stage/mode, and the switching between each stage:

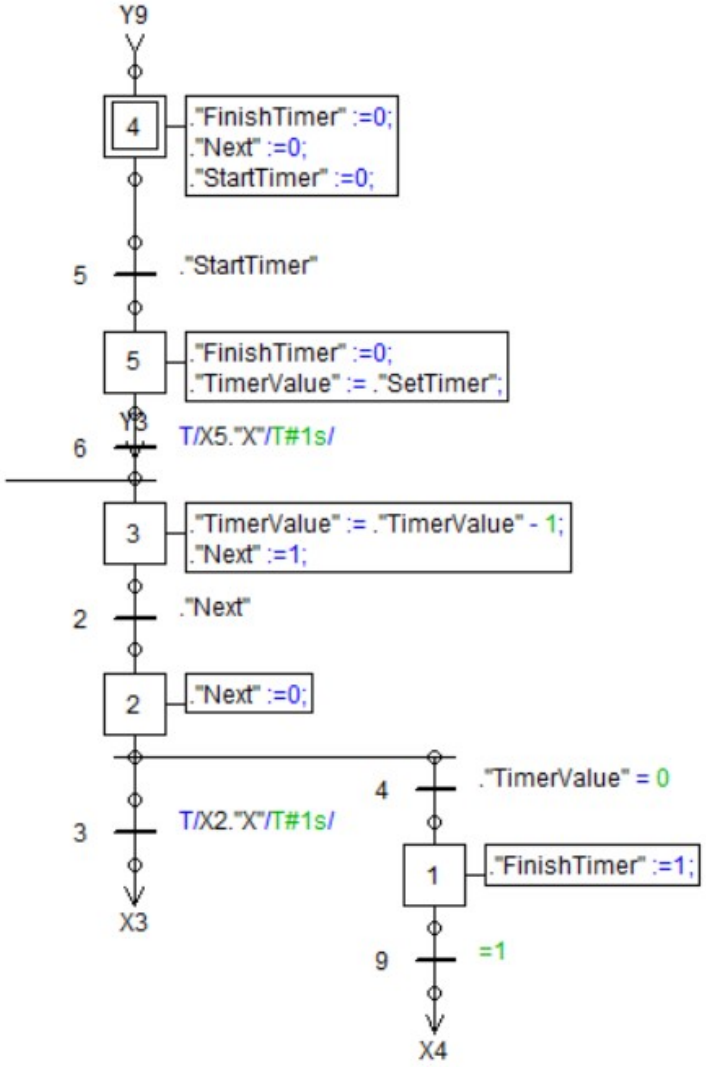
- On each mode-change, all the variables are reset to ensure that no previous state can affect the other (step 1 on grafcet)
- While a mode is maintained, the code loops in its relative part
- If the mode is changed before the end of a cycle, the system asynchronously resets to step 1



Timer

In order to change the timer period dynamically, we used a single timer of 1s with a sort of counter and some extra variables to monitor the state of the counter:

- **StartTimer:** Boolean Value set to True, when we need to start the timer and False otherwise
- **FinishTimer:** Boolean Value set to true to declare that the Timer is done.
- **SetTimer:** Integer set by the entity calling the timer
- **TimerValue:** monitors the actual count of the timer, triggers FinishTimer when it reaches 0.
- **Next:** Boolean value, required to ensure the correct flow of the logic



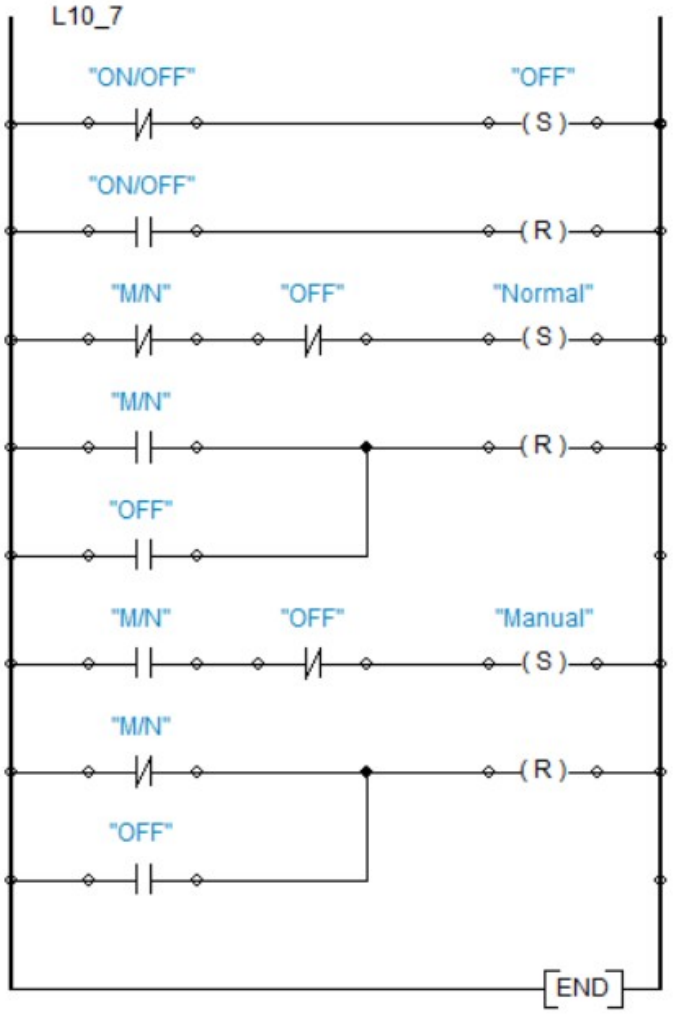
Mode Control

The Mode Control ladder code, as its name suggests, computes through some logic the actual mode of the system. It considers two switches ON/OFF and Manual/Normal:

1. If ON/OFF is set to OFF, then, whatever the state of the other switch is, Normal and Manual are reset and OFF is set.

2. If ON/OFF is set to ON, then:

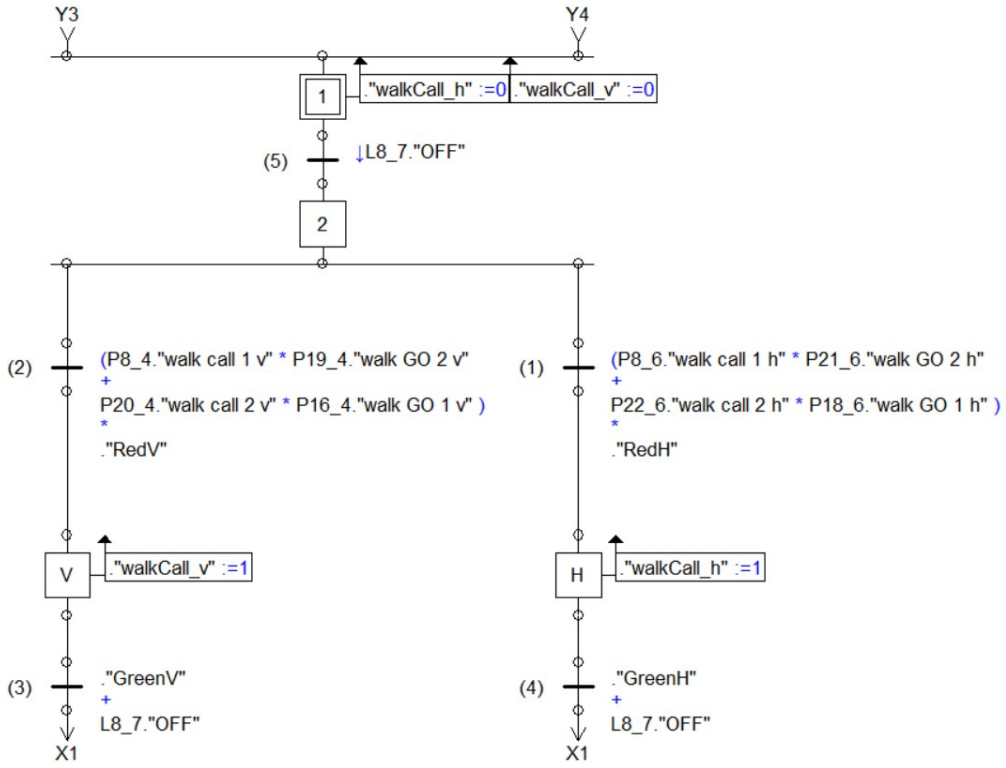
- If M/N is set to M, then OFF and Normal are reset, and Manual is set.
- If M/N is set to N, then OFF and Manual are reset, and Normal is set



Walk Call Logic

The walk call logic simulates the call button of the pedestrian when he arrives at a red traffic light. It has two calling variables (walkCall_v and walkCall_h), one for vertical lane, and the other for the horizontal lane.

Each variable is set when the person arrives to the cross intersection with a red light, then it is reset again when the relative traffic light switches to green.

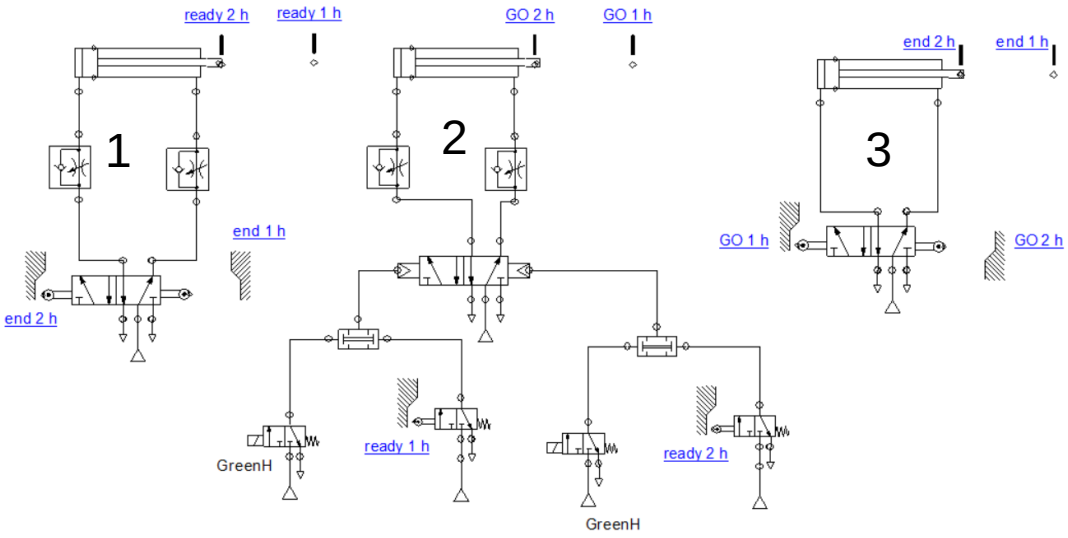


Simulation

In order to do the simulation we need some actuators to simulate the position of each moving element (cars and pedestrians). The logic works in the following way:

- The cylinder 1 lets the car/pedestrian arrive at the cross intersection. The arriving state is set by ready_1_h and ready_2_h for the forward and backward movement respectively
- Cylinder 2 is actuated in the following conditions:

1. 2+ if the object is moving forward (ready_1_h) and traffic light is green (GreenH)
2. 2- if the object is moving backward (ready_2_h) and traffic light is green (GreenH)

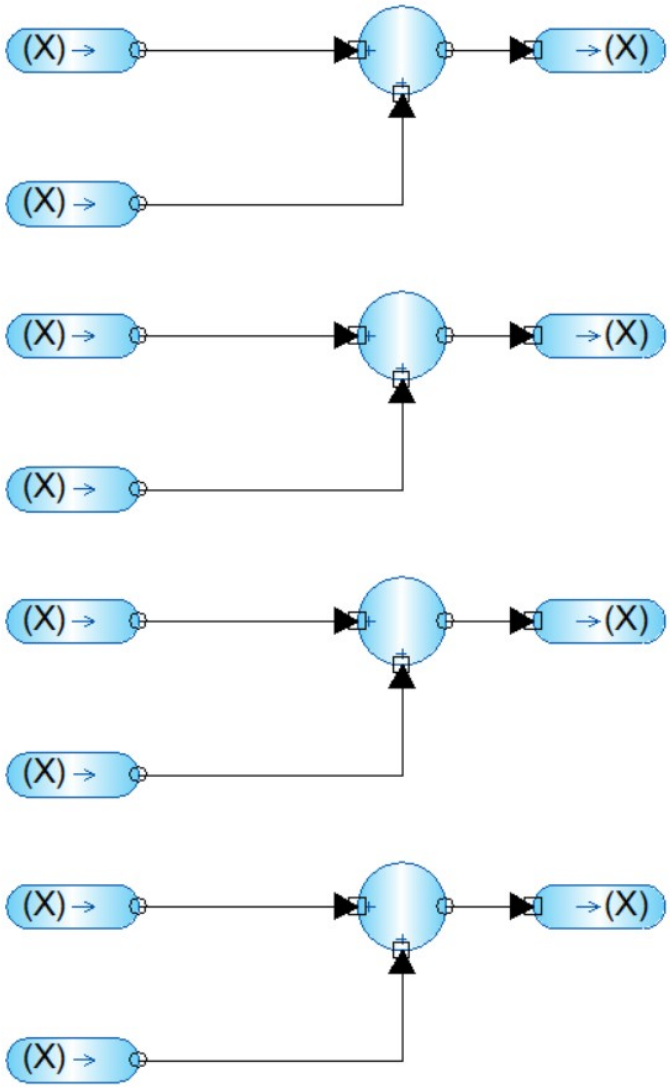


- Cylinder 3 has different roles:

1. For the car: it helps positioning and orienting the car when it is going forward or backward
2. For the pedestrian: it is used as a delay to start a new cycle, a normal timer is not used because we don’t need a specific time and make the simulation as random as possible

Simulation

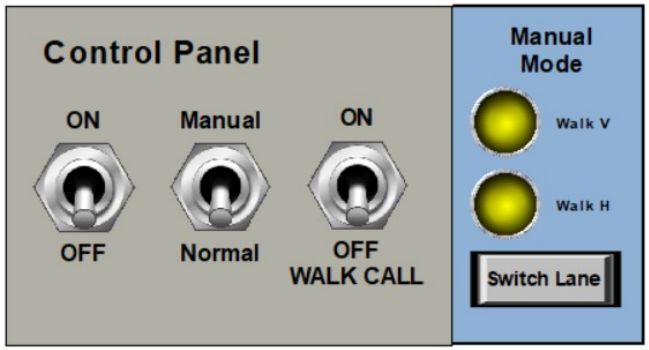
To compute the actual position having the extension of two cylinders, we simply use an add operation for each element (car_v, car_h, pedestrian_v, pedestrian_h)

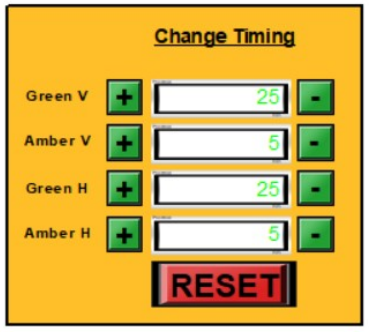


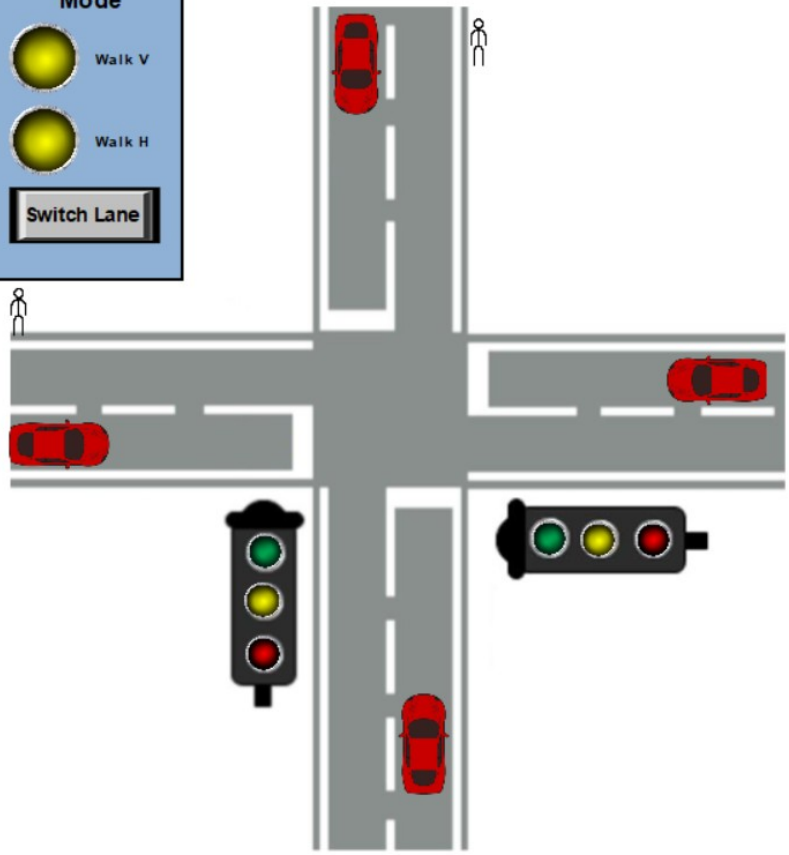
Simualtion

All the logic above is used by this control panel to process the simulation:

- ON/OFF and M/N switches to select the simulation mode
- Walk call switch enables or disables the pedestrian call in automatic mode when they arrive at the cross section with a red light.
- Walk V and Walk H LEDs are used to monitor the presence of waiting pedestrian
- Change Timing block is usable only in off mode, used to change the length of red and green signals



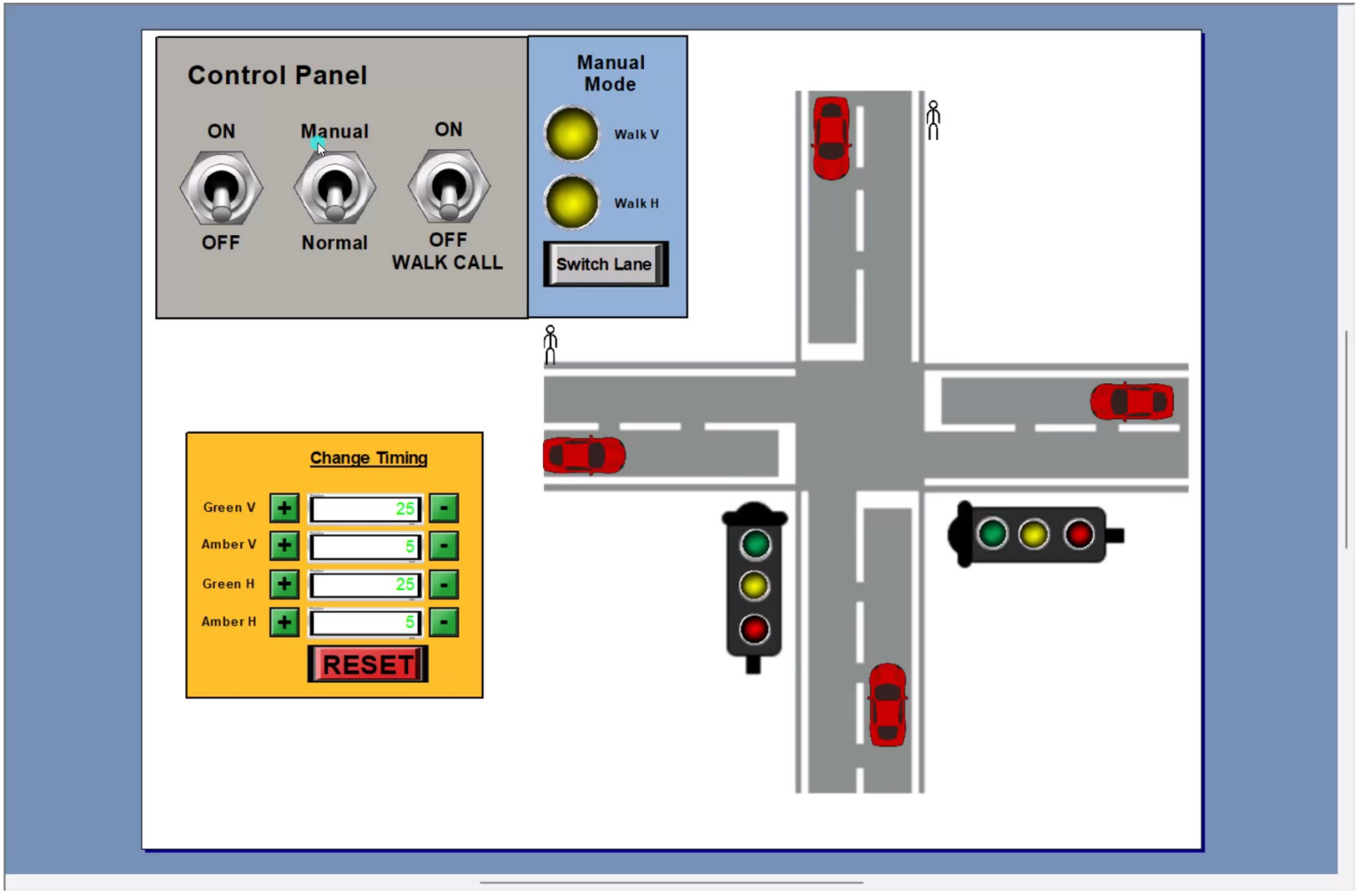




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Group17_Task1

Video recording

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Conclusions

The cross-intersection controller was designed and simulated by splitting the problem into small blocks (mode control, timer, pedestrian calls, simulation) that are combined at the end.

The system has three operating modes: Off, Normal and Manual.

The main sequence is written in Grafcet, while mode control, the timer and the pedestrian calls are written in ladder logic.

A single 1 s timer with a counter lets the operator change the green and red durations at runtime.

Pneumatic cylinders simulate the cars and pedestrians, which move only when their traffic light allows it.

Some computations run in parallel to the main program to keep the simulation flowing correctly.